

Baulkham Hills High School



YEAR 11

YEARLY EXAMINATION

2008

**MATHEMATICS
ADVANCED**

Time allowed: Three hours

DIRECTIONS TO CANDIDATES:

- **Approved calculators may be used**
- **Begin each question on a new page.**
- **Show all necessary working.**

Question 1

a) Factorise completely

1

i) $a^2 - 25$

1

ii) $x^2 - 3x + 2$

2

iii) $4m^3 - 32n^3$

b) Solve for x.

i) $6x^2 - x - 2 \geq 0$

2

ii) $|x + 2| = 4$

2

iii) $\frac{2x+3}{3} - \frac{3x-1}{4} = 3$

2

c) Evaluate correct to 2 sig. figures.

$$\sqrt[3]{\frac{86.54 \times 3.16}{2.6^3}}$$

2

Question 2

a) Convert $0.\dot{2}\dot{3}$ to a fraction in its simplest form.

2

b) Simplify $\frac{3^3}{(3^x)^3 \times 3^{-3x}}$

2

c) Express $\frac{3\sqrt{2}+1}{2\sqrt{3}-1}$ with a rational denominator.

2

d) If $(2\sqrt{3}-2)(\sqrt{3}-1) = a - \sqrt{b}$, find a and b.

2

e) At a shop all prices increased by 35%. One item now costs \$81.20. What is the old price of this item?

2

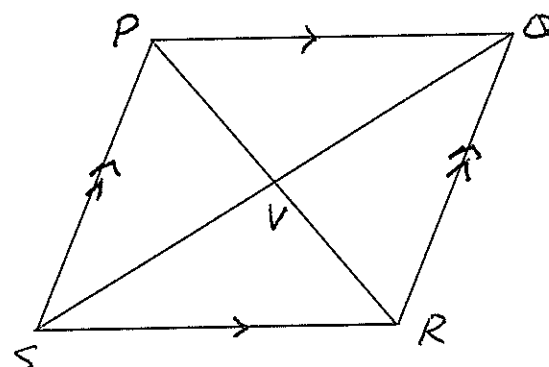
f) Express $(a^{\frac{1}{2}} - b^{\frac{-1}{2}})^2$ without indices.

2

Question 3

Marks

a)

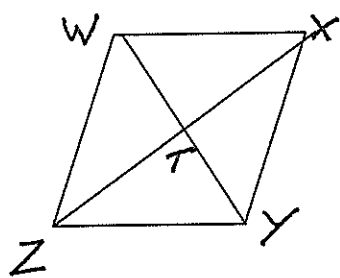


NOT TO SCALE

PQRS is a parallelogram whose diagonals intersect at V. Given that $\angle QPR = 68^\circ$ and $\angle QSR = 47^\circ$, find the size of $\angle SVR$. Give reasons for your answer.

2

b) In the diagram below, WXYZ is a rhombus

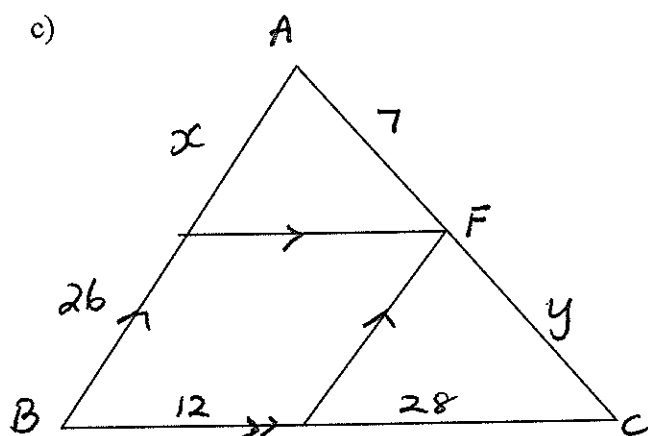


NOT TO SCALE

- Prove that $\triangle WXT \equiv \triangle YZT$
- State why $\angle WTX = 90^\circ$.

3
1

c)



NOT TO SCALE

In the diagram, BA is parallel to DF and EF parallel to BC. Find x and y (correct to 1 decimal place). Give reasons.

4

d) Calculate the size of each internal angle of a regular 20 sided polygon.

2

Question 4

7

a) Differentiate

i) $x^3 - 2x^2 + 1$

ii) $\sqrt{x^2 - 6}$

iii) $(2x + 1)^8$

iv) $\frac{5x^3 - x}{x^2}$

b) Simplify

6

i) $\frac{m^6 \times m^4}{m^8}$

ii) $\frac{m + 4}{m^2 + 2m - 8}$

iii) $\frac{3}{m^2 - 4} - \frac{5}{2m - 4}$

c) Consider the arithmetic sequence: $a, 3a - 1, 5a - 2$

1

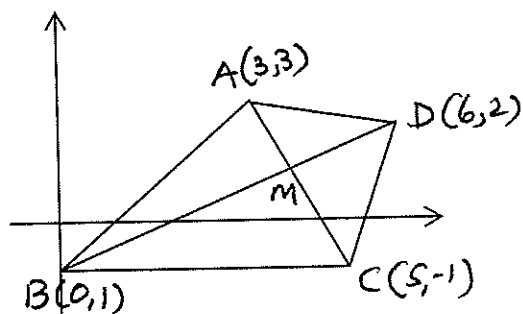
i) Find the common difference.

ii) Find the sum of the first twenty terms.

2

Question 5

In a quadrilateral ABCD, the points A, B, C and D are (3,3), (0,-1) and (5, -1) and (6,2) respectively. The line BD bisects the line AC at right angles at the point M



- Find the distance of BD.
- Show that the gradient of BD is $\frac{1}{2}$.
- Show that the equation of the line BD is $x - 2y - 2 = 0$.
- Show that the equation of the line AC is $2x + y - 9 = 0$.
- What are the coordinates of M?
- Find the area of ABCD.

1
1
2
3
2
3

Question 6

- If $f(x) = 2x^2 + 5x + 1$ calculate $f(-3)$.
- Consider the function $y = \sqrt{16 - x^2}$
 - State the domain.
 - Show that it is an even function.
- The equation $x^2 - 6x + y^2 - 12y + 20 = 0$ represents a circle.
 - Show that the centre is (3,6) and the radius is 5.
 - Show that the line $x - 12y + 1 = 0$ does not intersect or touch the circle.
- Show the region represented by $y < 4 - x^2$ and $y \geq 3^x - 3$.

2

1
1

2
2
4

Question 7

a) Sketch each of the following showing all important features:

9

i) $y = 3^x$

ii) $(x - 2)^2 + y^2 = 4$

iii) $y = \frac{-1}{x}$

iv) $y = |2x - 4|$

b) What is the range of $f(x) = \frac{1}{\sqrt{8-2x}}$

2

c) Find the exact value of

3

i) $\sin 330^\circ$

ii) $\cos 240^\circ$

iii) $\sec 225^\circ$

d) Find the equation of the tangent to the curve $y = 4 - x^2$ at $x = 1$.

2

QUESTION 8

a) If $90^\circ \leq \theta \leq 180^\circ$ and $\sin \theta = \frac{2}{3}$, find the exact value of $\cos \theta$.

2

b) Solve for θ where $0^\circ \leq \theta \leq 360^\circ$.

i) $\sin \theta = \frac{1}{\sqrt{2}}$

2

3

ii) $3 \tan^3 \theta = \tan \theta$

c) A person walks 10 km East from A to B and then 12 km South East to C.

i) Draw a neat diagram and mark the given information

2

ii) Find the distance to the nearest km between A and C.

2

iii) What is the bearing of A from C?

2

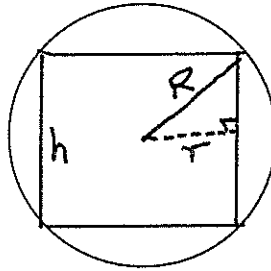
Question 9

- a) The population of kangaroos in a nature reserve is given by

$$P = 100 + 6t^2 - 2t^3 \quad (\text{where } t \text{ is in years})$$

- | | | |
|------|--|---|
| i) | find the stationary points of the function and determine their nature. | 4 |
| ii) | Find any point of inflexion. | 2 |
| iii) | Sketch the function in the domain $0 \leq t \leq 5$ | 2 |
| iv) | When was the population of the kangaroos increasing most rapidly. | 1 |

- b) A metal worker is required to cut a circular cylinder with radius r cm, height h cm, from a solid sphere of metal radius $R = 4$ cm. The diagram below shows a cross-section of the sphere and the cylinder.



- (i) Show that the volume $V \text{ cm}^3$ of the cylinder is given by

$$V = \frac{1}{4} \pi h (64 - h^2)$$

- (ii) Find the value of h such that the volume of the cylinder is maximum.

Question 10

- | | | |
|------|--|---|
| i) | For what values of m does the quadratic equation $x^2 + mx + (m+1)^2 = 0$ have two real roots. | 3 |
| ii) | If $a:b:c$ is $1:2:3$ and $a^2 + b^2 + c^2 = 504$ find the value of a , where a is a positive number | 2 |
| iii) | If α and β are the roots of equation $x^2 + 5x - 8 = 0$ find the value of $(\alpha + 2)(\beta + 2)$. | 2 |
| iv) | A series is defined by $s_n = 3n^2 - 11n$. Prove that the series s_n is an arithmetic progression. | 2 |
| v) | Consider the series $\cos^2 x + \cos^4 x + \cos^6 x + \dots$ $0^\circ < x < 90^\circ$ | 3 |
| | i) show that a limiting sum exists. | |
| | ii) Find the limiting sum expressing the answer in its simplest form. | |

END OF EXAMINATION

Question 1

a) (i) $(a-5)(a+5)✓$

(ii) $(x-2)(x-1)✓$

(iii) $4(m^3 - 8n^3)$

$= 4(m-2n)(m^2 + 2mn + 4n^2)✓$

b) (i) $6x^2 - 4x + 3x - 2 \geq 0$

$2x(3x-2) + 1(3x-2) \geq 0$

$(2x+1)(3x-2) \geq 0✓$

$x \leq -\frac{1}{2} \text{ or } x \geq \frac{2}{3}✓$

(ii) $x+2=4 \therefore x=2✓$

$-x-2=4 \quad x=-6✓$

(iii) $4(2x+3) - 3(3x-1) = 36✓$

$-x + 15 = 36$

$x = -21✓$

c) $= 0.125 \dots$

$\div 0.125 \text{ (5 sig figs)}$

$= 2.496479✓$

$\div 2.5✓$

(12)

Question 2

a) $x = 0.\dot{2}\dot{3}$

$100x = 23.2323 \dots✓$

$99x = 23$

$x = \frac{23}{99}✓$

b) $\frac{3^3}{3^0} = 27✓$

(c) $\frac{3\sqrt{2}+1}{2\sqrt{3}-1} \times \frac{2\sqrt{3}+1}{2\sqrt{3}+1}✓$

$= \frac{6\sqrt{6} + 2\sqrt{3} + 3\sqrt{2} + 1}{12-1}$

$= \frac{6\sqrt{6} + 2\sqrt{3} + 3\sqrt{2} + 1}{11}✓$

(d) $(2\sqrt{3}-2)(\sqrt{3}-1)$

$= 6 - 2\sqrt{3} - 2\sqrt{3} + 2$

$= 8 - 4\sqrt{3}✓$

$= 8 - \sqrt{48}$

$a=8 \quad b=48✓$

e) $\frac{81.20}{135} \times 100 = \$60.15✓$

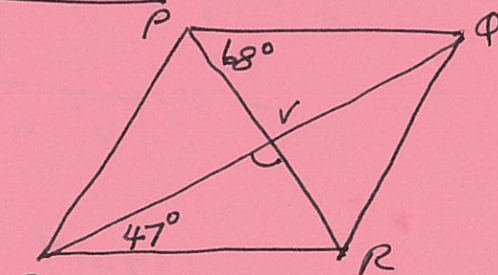
f) $(a^{\frac{1}{2}} - b^{\frac{1}{2}})^2 = a - 2a^{\frac{1}{2}}b^{\frac{1}{2}} + b✓$

$= a+b - 2\sqrt{\frac{a}{b}}✓$

(12)

Question 3

a)



$\angle PRS = 68^\circ$ [alternate $\angle$ m] lines

$\therefore \angle SVR = 180 - (47 + 68)$ [sum of Δ in 180°]

$= 65^\circ$

(b) In ΔWXT and ΔZTY

$\angle WTX = \angle ZTY$ (vert-opp $\angle$)✓

$WT = TY$

$TX = TS$

{ diagonals bisect each other }✓

$\Delta WTX \cong \Delta ZTY$ [SAS]✓

(c) Diagonals of a rhombus meet at 90° ✓

Question 3. (ctd)

$$\frac{y}{7} = \frac{28}{12} \checkmark \Rightarrow y = 16.3 \checkmark$$

$$\frac{x}{26} = \frac{7}{y} \checkmark \Rightarrow x = 11.1$$

Since the intercepts of the same set of $\parallel$ lines are in proportion. $\checkmark$

d) exterior $\angle = \frac{360}{20} = 18^\circ \checkmark$

$\therefore$ interior $\angle = 162^\circ \checkmark$ (12)

Question 4

i) $y = x^3 - 2x^2 + 1 \checkmark$

$$\frac{dy}{dx} = 3x^2 - 4x + 0.$$

ii) $y = \sqrt{x^2 - 6}$

$$y' = \frac{1}{2} (x^2 - 6)^{-\frac{1}{2}} \times 2x \checkmark$$

$$= \frac{x}{\sqrt{x^2 - 6}}$$

iii) $y = (2x+1)^8$

$$y' = 8(2x+1)^7 \times 2 \checkmark$$

$$= 16(2x+1)^7$$

(iv) $y = \frac{5x^3 - x}{x^2} = 5x - \frac{1}{x}$

$$y' = 5 + \frac{1}{x^2} \checkmark$$

Question 4 (ctd)

b)

i) $m^2 \checkmark$

ii) $\frac{m+4}{(m+4)(m-2)} = \frac{1}{m-2} \checkmark$

iii) $\frac{3}{(m-2)(m+2)} - \frac{5}{2(m-2)}$

$$= \frac{1}{(m-2)} \left[\frac{3}{m+2} - \frac{5}{2} \right]$$

$$= \frac{-4-5m}{2(m-2)(m+2)} \checkmark$$

c) Common difference

i) $= 3a - 1 - a = 2a - 1$
 $n = 20.$

ii) $S_{20} = \frac{20}{2} \{ 2a + 19(2a-1) \}$

$$= 20a + 380a - 190$$

$$= \underline{\underline{400a - 190}} \checkmark$$
 (16)

Question 5

$$\begin{aligned} \text{a) } d_{BD} &= \sqrt{(6-0)^2 + (2-1)^2} \\ &= \sqrt{36+9} \\ &= \sqrt{45} \checkmark \end{aligned}$$

$$\text{b) } m_{BD} = \frac{2-1}{6-0} = \frac{1}{6} \checkmark$$

$$\begin{aligned} \text{c) } y-2 &= \frac{1}{6}(x-6) \checkmark \\ 2y-4 &= x-6 \\ x-2y+2 &= 0. \end{aligned}$$

$$\begin{aligned} \text{d) } y+1 &= \frac{3+1}{3-5}(x-5) \\ y+1 &= \frac{4}{-2}(x-5) \\ y+1 &= -2x+10 \\ 2x+y-9 &= 0. \end{aligned}$$

$$\text{e) } M(4,1) \checkmark \checkmark$$

$$\text{f) } d_{AM} = \sqrt{(-1)^2 + (2)^2} = \sqrt{5} \checkmark$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} BD \times AM \times 2 \\ &= \sqrt{45} \times \sqrt{5} \\ &= \underline{\underline{15 \text{ unit}^2}} \checkmark \end{aligned}$$

(11)

Question 6.

$$\begin{aligned} \text{a) } f(-3) &= 2(-3)^2 + 5(-3) + 1 \checkmark \\ &= 18 - 15 + 1 \\ &= 4 \checkmark \end{aligned}$$

$$\text{b) (i) } -4 \leq x \leq 4 \checkmark$$

$$\begin{aligned} \text{(ii) } f(-x) &= \sqrt{16 - (-x)^2} \\ &= \sqrt{16 - x^2} \\ &= f(x) \checkmark \end{aligned}$$

Hence it is an even function.

$$\text{c) } x^2 - 6x + y^2 - 12y + 20 = 0.$$

$$\begin{aligned} \text{(i) } x^2 - 6x + 9 + y^2 - 12y + 36 + 20 &= 45 \\ (x-3)^2 + (y-6)^2 &= 25 \checkmark \\ &= 5^2 \end{aligned}$$

$\therefore$ Centre $(3,6)$ radius 5 .

$$\begin{aligned} x - 12y + 1 &= 0 \\ x &= 12y - 1 \end{aligned}$$

$$(12y-1-3)^2 + (y-6)^2 = 25$$

$$16(3y-1)^2 + (y-6)^2 = 25$$

$$16[9y^2 - 6y + 1] + y^2 - 12y + 36 = 25$$

$$145y^2 - 96y + 16 - 12y + 36 = 25$$

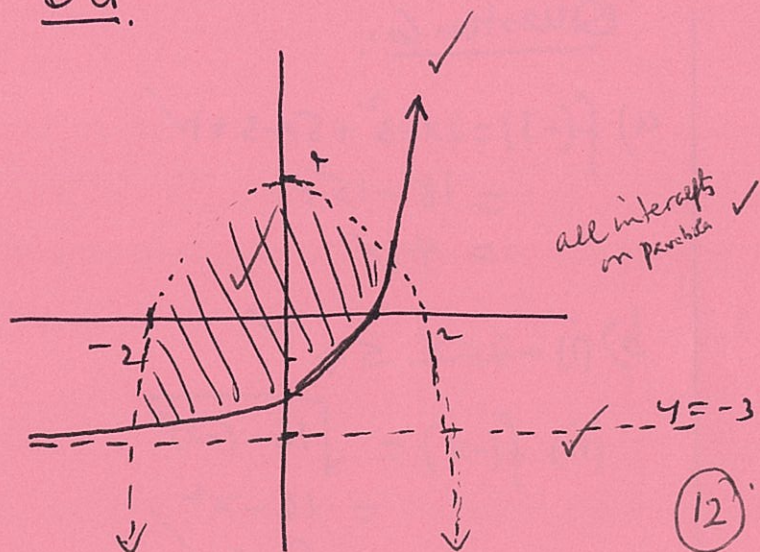
$$145y^2 - 108y + 27 = 0 \checkmark$$

$$\Delta = 108^2 - 4 \times 27 \times 145 < 0.$$

$\therefore$ There are no real roots.

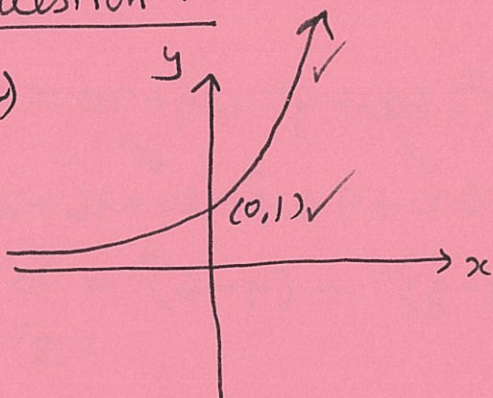
Hence $x - 12y + 1 = 0$ does not touch the circle.

6d.

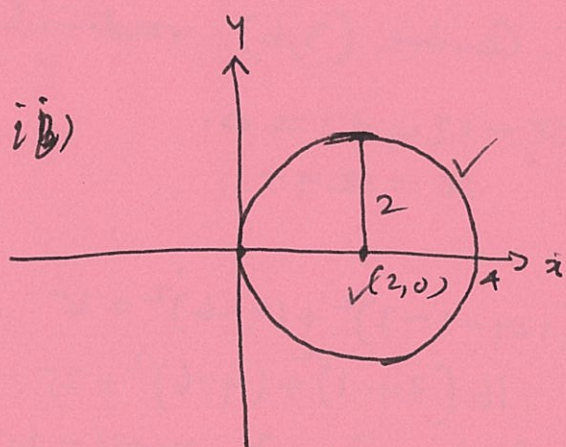


Question 7

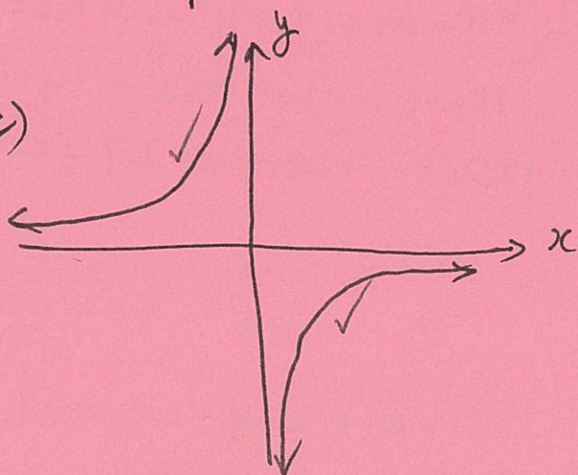
a) i)



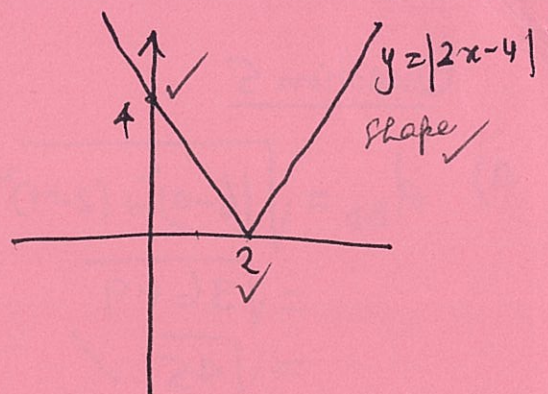
ii)



iii)



(20)



b) all real for x s.t. $f(x) > 0$

c) (i) $\sin(360 - 30^\circ) = -\frac{1}{2}$ ✓

(ii) $\cos(180 + 60^\circ) = -\frac{1}{2}$ ✓

(iii) $\sec 225^\circ = \frac{1}{\cos(180 + 45^\circ)} = -\sqrt{2}$ ✓

d) $y' = -2x$

$y'_{x=1} = -2$ ✓

$x = 1 \Rightarrow y = 4 - 1 = 3$

$y - 3 = -2(x - 1)$

$y + 2x - 5 = 0$

(15)

Toman.

VRSimB

Nada J.

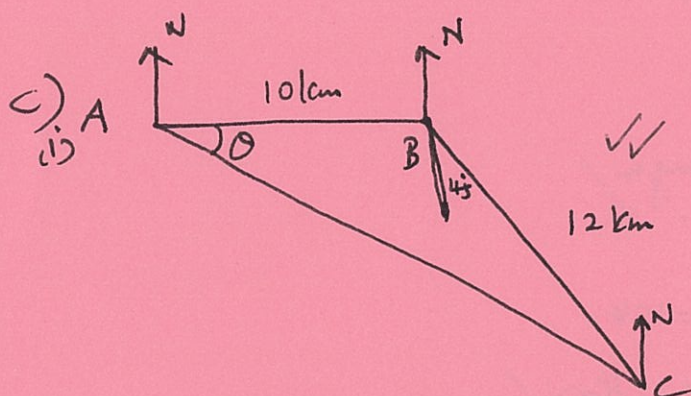
Question 8



$$\cos \theta = -\frac{\sqrt{5}}{3} \checkmark$$

b)(i)  $\sin \theta = \frac{1}{\sqrt{2}}$
 $\theta = 45^\circ, 135^\circ \checkmark$

(ii) $\tan \theta (3 \tan^2 \theta - 1) = 0$
 $\therefore \tan \theta = 0 \quad \tan \theta = \pm \frac{1}{\sqrt{3}} \checkmark$
 $\theta = 0^\circ, 180^\circ, 360^\circ \checkmark$
 $\theta = 30^\circ, 210^\circ, 150^\circ, 330^\circ \checkmark$



(ii) $AC^2 = 10^2 + 12^2 - 2 \times 10 \times 12 \cos 135^\circ \checkmark$
 $AC = 20 \text{ km.} \checkmark$

(iii) $\frac{12}{\sin \theta} = \frac{20}{\sin 135^\circ} \checkmark$
 $\sin \theta = \frac{12 \times \sin 135^\circ}{20}$
 $= 0.41717$
 $\theta = 24^\circ 39'$

Bearing $N 65^\circ 21' W$
 $291.029^\circ T \checkmark$

Question 9.

i) $P = 100 + 6t^2 - 2t^3$

$$P' = -6t^2 + 12t \quad P'' = -12t + 12.$$

Stationary pts occur when $P' = 0$.

$$-6t^2 + 12t = 0 \Rightarrow t = 0, t = 2$$

$$t = 0 \Rightarrow P = 100. \quad A(0, 100) \checkmark$$

$$t = 2 \Rightarrow P = 100 + 24 - 16 \quad B(2, 108) \checkmark$$

at $(0, 100)$ $P' = 0$ and $P'' > 0$

$\therefore (0, 100)$ minimum turning pt $\checkmark$

at $(2, 108)$ $P' = 0$ and $P'' < 0$.

$\therefore (2, 108)$ is a maximum turning pt. $\checkmark$

ii) When $P'' = 0 \Rightarrow t = 1$

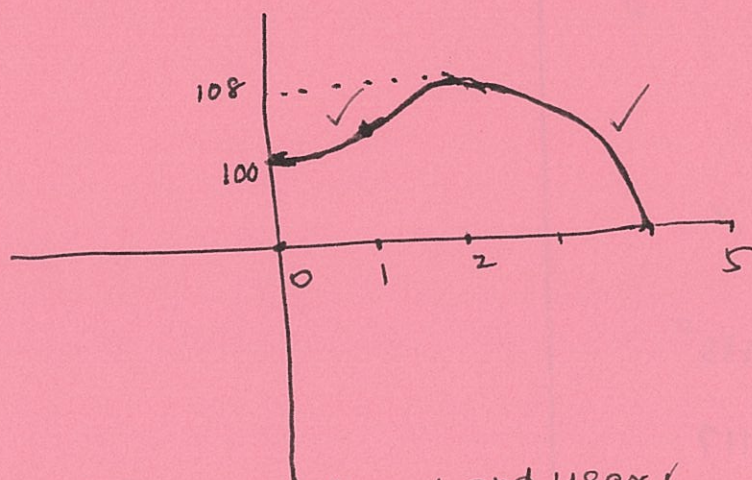
	1-	1	1+
P''	+ve	0	-ve

$\therefore$ There is a change in concavity. $\checkmark$

$$t = 1 \Rightarrow P = 100 + 6 - 2 = 104.$$

$\therefore (1, 104)$ is a point of inflexion $\checkmark$

(iii)



(iv) Between 1st and 2nd year $\checkmark$

Question 9

b).
(i) $V = \pi r^2 h$. — (2)

also $R^2 - r^2 = \left(\frac{h}{2}\right)^2 \checkmark$

$$16 - \frac{h^2}{4} = r^2 \text{ — (1)}$$

Sub (1) in (2).

$$\begin{aligned} V &= \pi h \cdot \left(16 - \frac{h^2}{4}\right) \checkmark \\ &= \frac{\pi h}{4} (64 - h^2). \end{aligned}$$

(ii) $V = 16\pi h - \frac{\pi h^3}{4}$.

$$\frac{dV}{dh} = 16\pi - \frac{3\pi h^2}{4} \checkmark$$

For a max volume
 $\frac{dV}{dh} = 0$ and $\frac{d^2V}{dh^2} < 0$. $\checkmark$

$$64\pi - 3\pi h^2 = 0$$

$$h^2 = \frac{64}{3}$$

$$h = \frac{8}{\sqrt{3}} \checkmark$$

$$\frac{d^2V}{dh^2} = 0 - \frac{6\pi}{4} \cdot h. \text{ This is always } < 0. \text{ for all } h.$$

$\therefore$ Value for h for max Volume

$$\text{is } \frac{8}{\sqrt{3}} \text{ or } \frac{8\sqrt{3}}{3} \text{ cm} //$$

Question 10.

i) $a=1$ $b=m$ $c=(m+1)^2$

For real roots $\Delta \geq 0$ ✓

$$m^2 - 4 \times 1 (m+1)^2 \geq 0.$$

$$m^2 - 2^2(m+1)^2 \geq 0$$

$$[m - 2(m+1)][m + 2(m+1)] \geq 0.$$

$$(-m-2)(3m+2) \geq 0.$$

$$(m+2)(3m+2) \leq 0 \checkmark$$

$$-2 \leq m \leq -\frac{2}{3} \checkmark$$

ii) $1a : 2a : 3a.$

$$a^2 + (2a)^2 + (3a)^2 = 504 \checkmark$$

$$14a^2 = 504$$

$$a^2 = 36$$

$$a = 6 \checkmark$$

iii) $\alpha + \beta = -5$, $\alpha\beta = -8$

$$(\alpha+2)(\beta+2) = \alpha\beta + 2(\alpha+\beta) + 4.$$

$$= -8 + 2(-5) + 4$$

$$= \underline{\underline{-14}} \checkmark$$

iv) $S_1 = -8$, $S_2 = -10$, $S_3 = 27 - 53 = -6$

$$\checkmark \begin{cases} T_1 = -8 \\ T_2 = -2 \\ T_3 = 4 \end{cases} \begin{matrix} \uparrow +6 \\ \uparrow +6 \end{matrix}$$

There exists a common diff.
 $\therefore S_n$ defines an AP.

(iv) $\sin^2 x + \sin^4 x + \sin^6 x + \dots$

$$\frac{T_2}{T_1} = \frac{T_3}{T_2} = \sin^2 x.$$

$\therefore$ there exist a common ratio $\sin^2 x$.

$$\forall 0 < x < 90 \quad 0 < \sin^2 x < 1$$

$$\text{i.e. } \sin^2 x < 1$$

$$\therefore S_x = \frac{a}{1-r} = \frac{\sin^2 x}{1-\sin^2 x}$$

$$= \frac{\sin^2 x}{\cos^2 x}$$

$$= \tan^2 x.$$

$\forall 0 < x < 90$ $\tan^2 x$ is defined hence exist.

(v) ii) $\cos^2 x + \cos^4 x + \cos^6 x + \dots$

Common ratio $\frac{\cos^4 x}{\cos^2 x} = \cos^2 x \checkmark$

$$\forall x < x < 90 \quad -1 < \cos^2 x < 1$$

$$\therefore \cos^2 x < 1 \checkmark$$

Hence the common ratio < 1

$$\therefore S_x = \frac{a}{1-r}$$

$$= \frac{\cos^2 x}{1-\cos^2 x} = \frac{\cos^2 x}{\sin^2 x}$$

$$= \underline{\underline{\cot^2 x}} \checkmark$$

(12)